Problem 1.) - Waveguides

Part A.)

Using Eq. (2.1.7), estimate the number of modes that can be supported in a planar dielectric waveguide that is 220 μm wide, and has n_1 = 1.572 and n_2 = 1.537 at the free-space source wavelength (λ), which is 1.5 μm Compare your estimate with formula (2.1.11) in which Int(x) is the integer function; it drops the decimal fraction of (x).

Part B.)

A single-mode optical fiber has a core diameter of 7.8 um, n_2 = 1.4372, Δ = 0.392 %. What is the waveguide dispersion coefficient at 1308.7 nm

Note: Please use all given values in your calculations (the answer will be sensitive to given values). Pay attention to the values given in each part of the problem, as they are particular to that part.

Problem 2.) - Single-Mode Fibers

Part A.)

What should be the core radius of a single-mode fiber that has a core of n_1 = 1.3723, cladding of n_2 = 1.3657, and is to be used with a source wavelength of 1.579 um ? What is the V-number of a fiber that has a diameter of 8.345 um and is operating at 1.579 um ?

Part B.)

A different single-mode optical fiber has a core diameter of 8.345 um and a core refractive index 1.372. The normalized index difference is 0.325%. The cladding diameter is 127.2 um . Calculate the numerical aperture and the acceptance angle of the fiber. What is the single-mode cutoff frequency (λ_c) of the fiber?

Note: Please use all given values in your calculations (the answer will be sensitive to given values). Pay attention to the values given in each part of the problem, as they are particular to that part.

Problem 3.) - Multi-Mode Fibers

Part A.)

Calculate the number of allowed modes in a multimode step-index fiber that has a core of refractive index of 1.4735 and diameter 127.3 μm, and a cladding of refractive index of 1.4532 if the source wavelength is 857 nm.

Part B.)

A step-index fiber has a core diameter of 127 μm and a refractive index of 1.4600. The cladding has a refractive index of 1.4400. Calculate the numerical aperture of the fiber, acceptance angle from air, and the number of modes sustained when the source wavelength is 857 nm.

Note: Please use all given values in your calculations (the answer will be sensitive to given values). Pay attention to the values given in each part of the problem, as they are particular to that part.

ECE 4520/5520 HW 3 Solutions

Problem (1)

Part A.)

Using Eq. (2.1.7), estimate the number of modes that can be supported in a planar dielectric waveguide that is 220 μm wide, and has n_1 = 1.572 and n_2 = 1.537 at the free-space source wavelength (λ), which is 1.5 μm Compare your estimate with formula (2.1.11) in which Int(x) is the integer function; it drops the decimal fraction of (x).

Part B.)

A single-mode optical fiber has a core diameter of 7.8 um, n_2 = 1.4372, Δ = 0.392 %. What is the waveguide dispersion coefficient at 1308.7 nm

Note: Please use all given values in your calculations (the answer will be sensitive to given values). Pay attention to the values given in each part of the problem, as they are particular to that part.

Part (A.)

$$\text{width} = 220 \times 10^{-6}\ m$$

$$n_1 = 1.572$$

$$n_2 = 1.537$$

$$\lambda = 1.5 \times 10^{-6}$$

Step (1) - Find V#

$$V\# = \left[\frac{\left(2\pi * \left(\frac{\text{width}}{2}\right)\right)}{\lambda}\right] * \left[\left(n_1^2 - n_2^2\right)^{1/2}\right]$$

$$V\# = (460.7669) * (0.32987)$$

$$V\# = 151.9937$$

Problem ① ct.)

Part Ⓐ ct.)

Step 2.) # Modes

$$\# \text{Modes} = \text{Int}\left(\frac{(2)(V\#)}{\pi}\right) + 1$$

$$\# \text{Modes} = 97.7622$$

$$\boxed{\# \text{Modes} = 97}$$

Part Ⓑ:

diameter $= 7.8 \times 10^{-6}$ m

$n_2 = 1.4372$

$\Delta = 0.392\%$

$\lambda = 1308.7 \times 10^{-9}$

Step 1.) Apply Equations (2.5.6 / 2.5.7)

$$D_w = -\frac{83.76\,(\lambda)\,(\mu m)}{[a\,(\mu m)]^2\, n_2} \approx \frac{(-83.76) * (1.3087 \times 10^{-6})}{(3.9\,(10^{-6})^2)(1.4872)}$$

$$D_w = \frac{-1.096167 \times 10^{-4}}{2.18598 \times 10^{-11}} \rightarrow D_w =$$

$$D_w = -5014531.14 \approx -5.0145 \times 10^{6}$$

$$\boxed{\text{Waveguide Dispersion Coefficient} = -5.0145 \times 10^{6}}$$

ECE 4520/5520 HW 3 Solutions

Problem (2)

Part A.)

What should be the core radius of a single-mode fiber that has a core of n_1 = 1.3723, cladding of n_2 = 1.3657, and is to be used with a source wavelength of 1.579 um? What is the V-number of a fiber that has a diameter of 8.345 um and is operating at 1.579 um?

Part B.)

A different single-mode optical fiber has a core diameter of 8.345 um and a core refractive index 1.372. The normalized index difference is 0.325%. The cladding diameter is 127.2 um. Calculate the numerical aperture and the acceptance angle of the fiber. What is the single-mode cutoff frequency (λ_c) of the fiber?

Note: Please use all given values in your calculations (the answer will be sensitive to given values). Pay attention to the values given in each part of the problem, as they are particular to that part.

Part (A)

$n_1 = 1.3723$

$n_2 = 1.3657$

$\lambda = 1.579 \times 10^{-6}$

Step 1.) Core Radius

$$V = \left(\frac{2\pi a}{\lambda}\right)(n_1^2 - n_2^2)^{1/2} \leq 2.405 \rightarrow$$

using this... we can solve for the bound of (a)

$$a \leq \frac{(2.405 * \text{wavelength})}{(n_1^2 - n_2^2)^{1/2}\, 2\pi}$$

$a \leq 4.496 \times 10^{-6}\, m$

Found that $a \leq 4.496 \times 10^{-6}\, m$ or $d \leq 8.992 \times 10^{-6}$

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Problem (2) ct.) Part (A) ct.)

Step 2.) Find V# Using Given Values

$$d = 8.345 \times 10^{-6}$$

$$\lambda = 1.579 \times 10^{-6}$$

$$V = \left(\frac{(2\pi a)}{\lambda}\right)\left(n_1^2 - n_2^2\right)^{1/2}$$

$$V = \left[\frac{(2\pi)\left(\frac{8.345 \times 10^{-6}}{2}\right)}{1.579 \times 10^{-6}}\right]\left[(1.3723)^2 - (1.3657)^2\right]^{1/2}$$

$$V = (16.60329)(0.13443)$$

$$V\# = 2.231941$$

Found that $a \leq 4.496 \times 10^{-6}$ m or $d \leq 8.992 \times 10^{-6}$

Found that $V\# = 2.231941$

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Problem ② ct.)

Part (B.)

$d = 8.345 \times 10^{-6}$

$n_1 = 1.372$

$\Delta = 0.325\% = 0.00325$

$Cd = 127.2 \times 10^{-6}$

Step 1.) Find Numerical Aperture

$$NA = n_1 (2\Delta)^{1/2} \rightarrow (1.372)((2)\,0.00325)^{1/2}$$

$$NA = 0.110614$$

Step 2.) Acceptance Angle

$$\alpha_{max} = \sin^{-1}\left(NA/n_0\right) \rightarrow n_0 = 1$$

$$\alpha_{max} = 6.3507°$$

Step 3.) Single-Mode Cutoff

$$\lambda_c = \frac{(2\pi a\, NA)}{2.405} \rightarrow \lambda_c = \frac{\left(2\pi\left(\frac{8.345\times10^{-6}}{2}\right)0.110614\right)}{2.405}$$

$$\lambda_c = 1.20579 \times 10^{-6}\ m$$

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Problem ② ct.)

Part Ⓑ.

$$\boxed{\begin{aligned} &\text{Numerical Aperture } (NA) = 0.110614 \\ &\text{Acceptance Angle } (\alpha_{max}) = 6.3507^\circ \\ &\text{Cutoff Frequency } (\lambda_c) = 1.20579 \times 10^{-6}\,m \end{aligned}}$$

ECE 4520/5520 HW 3 Solutions

Problem (3)

Part A.)

Calculate the number of allowed modes in a multimode step-index fiber that has a core of refractive index of 1.4735 and diameter 127.3 μm, and a cladding of refractive index of 1.4532 if the source wavelength is 857 nm.

Part B.)

A step-index fiber has a core diameter of 127 μm and a refractive index of 1.4600. The cladding has a refractive index of 1.4400. Calculate the numerical aperture of the fiber, acceptance angle from air, and the number of modes sustained when the source wavelength is 857 nm.

Note: Please use all given values in your calculations (the answer will be sensitive to given values). Pay attention to the values given in each part of the problem, as they are particular to that part.

Part (A)

$n_1 = 1.4735$

$d = 127.3 \times 10^{-6}\ m$

$n_2 = 1.4532$

$\lambda = 857 \times 10^{-9}$

Step 1.) Find V#

$$V\# = \left(\frac{2\pi a}{\lambda}\right)\left(n_1^2 - n_2^2\right)^{1/2}$$

$$V\# = \left[\frac{(2\pi)\left(\frac{127.3\times10^{-6}}{2}\right)}{(857\times10^{-9})}\right] * \left[1.4735^2 - 1.4532^2\right]^{1/2}$$

ECE 4520/5520 HW 3 Solutions

Problem (3) ct.) Part (A) ct.)

$V\# = (466.6566) * (0.24375)$

$V\# = 113.7475$

Step 2.) Modes

$\# \text{ Modes} = \frac{V^2}{2} \rightarrow \# \text{Modes} = 6469.24688$

$$\boxed{\begin{gathered}\text{V-number of Multi-Mode Fiber} = 113.7475 \\ \text{Number of Modes of Multi-Mode Fiber} = 6469.2469\end{gathered}}$$

Part (B)

$d = 127 \times 10^{-6}$ m

$n_1 = 1.4600$

$n_2 = 1.4400$

$\lambda = 857 \times 10^{-9}$

Step 1.) Numerical Aperture

$NA = (n_1^2 - n_2^2)^{1/2} = (1.46^2 - 1.44^2)^{1/2}$

$NA = 0.240831$

ECE 4520/5520 HW 3 Solutions

Problem ③ ct.) Part Ⓑ ct.)

Step 2.) Acceptance Angle

$$\alpha_{max} = \sin^{-1}\left(NA/n_0\right) \qquad n_0 = 1$$

$$\alpha_{max} = 13.9356^{\circ}$$

Step 3.) # modes

$$V\# = \left(\frac{2\pi a}{\lambda}\right) NA = \left(\frac{(2\pi)\left(\frac{127\times10^{-6}}{2}\right)}{857\times10^{-9}}\right)(0.240831)$$

$$V\# = (465.5569)(0.240831)$$

$$V\# = 112.1205$$

$$\#\text{ Modes} = \#V^2/2 \rightarrow \#\text{ mods} = \frac{(112.1205)^2}{2}$$

$$\#\text{ modes} = 6285.507162$$

$$\boxed{\begin{array}{ll} \text{Numerical Aperture (NA)} & = 0.240831 \\ \text{Acceptance Angle } (\alpha_{max}) & = 13.956^{\circ} \\ \#\text{ Modes In Fiber} & = 6285.507162 \end{array}}$$

```
% ECE 4520/5520 Homework 4 Solutions
% Due: April 21, 2023

clear all;
clc;

% Problem 1: Waveguides

% Part A.) V-Number and Number of Modes
fprintf↙
('----------------------------------------------------------------------------------------↙
----------------- \n');
fprintf('\nProblem 1 Part A.) \n \n');

% Define Constants

width_prob1_a = 220.*10.^-6;
index_n1_prob1_a = 1.572;
index_n2_prob1_a = 1.537;
wavelength_prob1_a = 1.5.*10.^-6;

% Step 1 - Calculating V-Number
% Using Equation 2.1.7
V_prob1_a = (((2.*pi).*(width_prob1_a./2))./(wavelength_prob1_a)).*(((index_n1_prob1_a.↙
^2)-(index_n2_prob1_a.^2)).^0.5);

% Step 2 - Calculate M ... number of modes
M_prob1_a = floor(((2).*(V_prob1_a))./(pi)) + 1;
fprintf('The number of modes supported in the waveguide is: %d\n', M_prob1_a);

% Part B.) Waveguide Dispersion
fprintf('\nProblem 1 Part B.) \n \n');

% Define Constants
diameter_prob1_b = 7.8.*10.^-6;
index_n2_prob1_b = 1.4372;
delta_prob1_b = 0.00392;
wavelength_prob1_b = 1308.7*10.^-9;

% Step 1.) Apply Eq. 2.5.6/2.5.7

Dw_prob1_b = (-1).*((83.76.*wavelength_prob1_b)./(((diameter_prob1_b./2).^2).↙
*index_n2_prob1_b));
fprintf('The dispersion of the waveguide is: %d\n', Dw_prob1_b);
fprintf↙
('\n--------------------------------------------------------------------------------------↙
------------------- \n');


% Problem 2: Single-Mode Fibers
% Part A.) Single-Mode Fibers - Core Radius and V #
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```
fprintf('\nProblem 2 Part A.) \n \n');

% Define Constants
index_n1_prob2_a = 1.3723;
index_n2_prob2_a = 1.3657;
wavelength_prob2_a = 1.579.*10.^-6;
diamter_prob2_a = 8.345.*10.^-6;
wavelength_op_prob2_a = 1.579.*10.^-6;

% Step 1.) Solve for core radius ... and therefore core diameter
% When V is greater than or equal to 2.405 ...

core_radius_prob2_a_term_1 = (2.405.*wavelength_prob2_a);
core_radius_prob2_a_term_2 = (((index_n1_prob2_a).^2) - ((index_n2_prob2_a).^2)).^0.5;
core_radius_prob2_a_term_3 = (2.*pi);
core_radius_prob2_a = (core_radius_prob2_a_term_1)./((core_radius_prob2_a_term_2).*↙
(core_radius_prob2_a_term_3));
core_diameter_prob2_a = core_radius_prob2_a.*2;
fprintf('The core radius of the single mode fiber is: %d\n', core_radius_prob2_a);
fprintf('The core diameter of the single mode fiber is: %d\n', core_diameter_prob2_a);

% Step 2.) What is the V Number of a fiber with diameter 8.345 um and cuttof
% 1.579 um
core_diameter_prob2_a_second_part = 8.345.*10.^-6;
v_number_prob2_a_term_1 = (2.*pi.*(core_diameter_prob2_a_second_part./2))./↙
(wavelength_prob2_a);
v_number_prob2_a_term_2 = core_radius_prob2_a_term_2;
v_number_prob2_a = v_number_prob2_a_term_1.*v_number_prob2_a_term_2;
fprintf('The V-Number of the single mode fiber is: %d\n', v_number_prob2_a);

% Part B.) Single-Mode Fiber
fprintf('\nProblem 2 Part B.) \n \n');

% Define Constants
core_diameter_prob2_b = 8.345.*10.^-6;
index_n1_prob2_b = 1.372;
normalized_index_difference_prob2_b = 0.00325;
cladding_prob2_b = 127.2.*10.^-6;

% Step 1.) Find the numerical aperture of the fiber
num_ap_prob2_b = (index_n1_prob2_b).*((2.*normalized_index_difference_prob2_b).^0.5);
fprintf('The Numerical Aperture of the single mode fiber is: %d\n', num_ap_prob2_b);

% Step 2.) Find the Acceptance Angle of the fiber
no_prob2_b = 1;
alpha_max_prob2_b = asind(num_ap_prob2_b./no_prob2_b);
fprintf('The Acceptance Angle (in degrees) of the single mode fiber is: %d\n',↙
alpha_max_prob2_b);

% Step 3.) What is teh single-mode cutoff frequency of the fiber?
lambda_c_prob2_b = (2.*pi.*(core_diameter_prob2_b./2).*num_ap_prob2_b)./(2.405);
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fprintf('The Single-Mode Cutoff Frequency of the single mode fiber is: %d\n',
lambda_c_prob2_b);
fprintf
('\n-------------------------------------------------------------------------------------
------------------- \n');

% Problem 3: Multi-Mode Fibers
% Part A.) Multi-Mode Fibers - Number of Modes
fprintf('\nProblem 3 Part A.) \n \n');

% Define Constants
index_n1_prob3_a = 1.4735;
index_n2_prob3_a = 1.4532;
core_diameter_prob3_a = 127.3.*10.^-6;
wavelength_prob3_a = 857.*10.^-9;

% Step 1.) Find V-Number
v_number_prob3_a_term1 = (2.*pi.*(core_diameter_prob3_a./2))./(wavelength_prob3_a);
v_number_prob3_a_term2 = (((index_n1_prob3_a).^2) - ((index_n2_prob3_a).^2)).^0.5;
v_number_prob3_a = v_number_prob3_a_term1.*v_number_prob3_a_term2;
fprintf('The V-Number of the Multi-Mode Fiber is: %d\n', v_number_prob3_a);

% Step 2.) Find the number of Modes
modes_prob3_a = (v_number_prob3_a.^2)./2;
fprintf('The number of modes of the Multi-Mode Fiber is: %d\n', modes_prob3_a);

% Part B.) Multi-Mode Fiber  NA, Acceptance Angle, Number of Modes
fprintf('\nProblem 3 Part B.) \n \n');

% Define Constants
core_diameter_prob3_b = 127.*10.^-6;
index_n1_prob3_b = 1.4600;
index_n2_prob3_b = 1.4400;
wavelength_prob3_b = 857.*10.^-9;

% Step 1.) Find the Numerical Aperture
num_ap_prob3_b = (((index_n1_prob3_b).^2) - ((index_n2_prob3_b).^2)).^0.5;
fprintf('The Numerical Aperture of the Multi-Mode Fiber is: %d\n', num_ap_prob3_b);

% Step 2.) Find the Acceptance Angle
no_prob3_b = 1;
alpha_max_prob3_b = asind(num_ap_prob3_b./no_prob3_b);
fprintf('The Acceptance Angle (in degrees) of the Multi-Mode fiber is: %d\n',
alpha_max_prob3_b);

% Step 3.) Find the Number of Modes
v_number_prob3_b = (2.*pi.*(core_diameter_prob3_b./2))./(wavelength_prob3_b).*
(num_ap_prob3_b);
modes_prob3_b = (v_number_prob3_b.^2)./2;
fprintf('The number of modes of the Multi-Mode Fiber is: %d\n', modes_prob3_b);
fprintf
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('\n--------------------------------------------------------------------------------------
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Problem 1 Part A.)

The number of modes supported in the waveguide is: 97

Problem 1 Part B.)

The dispersion of the waveguide is: -5.014531e+06

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Problem 2 Part A.)

The core radius of the single mode fiber is: 4.496024e-06
The core diameter of the single mode fiber is: 8.992049e-06
The V-Number of the single mode fiber is: 2.231941e+00

Problem 2 Part B.)

The Numerical Aperture of the single mode fiber is: 1.106142e-01
The Acceptance Angle (in degrees) of the single mode fiber is: 6.350721e+00
The Single-Mode Cutoff Frequency of the single mode fiber is: 1.205791e-06

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Problem 3 Part A.)

The V-Number of the Multi-Mode Fiber is: 1.137456e+02
The number of modes of the Multi-Mode Fiber is: 6.469030e+03

Problem 3 Part B.)

The Numerical Aperture of the Multi-Mode Fiber is: 2.408319e-01
The Acceptance Angle (in degrees) of the Multi-Mode fiber is: 1.393564e+01
The number of modes of the Multi-Mode Fiber is: 6.285554e+03

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